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SUPERSEDING
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31 August 1978

MILITARY SPECIFICATION

TEST STAND, AIRCRAFT PNEUMATIC COMPONENTS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirement for a test stand capable of testing pneumatic and electro-pneumatic components up to 5000 pounds-force per square inch.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

DD-G-451	Glass, Float or Plate, Sheet, Figured (Flat, for Glazing Mirrors and Other Uses)
GG-G-76	Gauges, Pressure and Vacuum, Dial Indicating (For Air, Steam, Oil, Water, Ammonia and Chloro-Fluoro Hydrocarbon Gases and Compressed Gases)
QQ-C-320	Chromium Plating (Electrodeposited)
TT-L-32	Lacquer, Cellulose Nitrate, Gloss for Aircraft Use

MILITARY

MIL-C-104	Crates, Wood, Lumber and Plywood Sheathed, Nailed and Bolted
MIL-P-116	Preservation, Methods of
MIL-B-121	Barrier Material, Greaseproofed, Waterproofed, Flexible

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: the Engineering Division, San Antonio ALC/MMEDO, Kelly AFB, Texas 78241 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 4920

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MIL-T-152	Treatment, Moisture- and Fungus-Resistant, of Communications, Electronic, and Associated Electrical Equipment
MIL-P-5518	Pneumatic Systems, Aircraft, Design, Installation and Data Requirements for
MIL-T-6845	Tubing, Steel, Corrosion-Resistant (304), Aerospace Vehicle Hydraulic System 1/8 Hard Condition
MIL-W-6858	Welding, Resistance, Spot and Seam
MIL-T-7081	Tube, Aluminum Alloy, Seamless, Round, Drawn, 6061 Aircraft Hydraulic Quality
MIL-P-7105	Pipe Threads, Taper, Aeronautical National Form, Symbol ANPT, General Requirements for
MIL-S-7742	Screw Threads, Standard, Optimum Selected Series, General Specification for
MIL-A-8421	Air Transportability Requirements, General Specification for
MIL-W-8604	Welding of Aluminum Alloys, Process for
MIL-W-8611	Welding, Metal Arc and Gas, Steels, and Corrosion and Heat Resistant Alloys, Process for
MIL-P-15024	Plates, Tags And Bands for Identification of Equipment
MIL-T-22085	Tape, Pressure-sensitive Adhesive, Preservation and Sealing
MIL-C-83488	Coating Aluminum, Ion Vapor Deposited
MIL-C-87115	Coating, Immersion Zinc/Chromate Dispersion

STANDARDS

FEDERAL

FED-STD-595	Colors

MILITARY

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-143	Standards and Specifications, Order of Precedence for the Selection of
MIL-STD-808	Finishes, Protective and Codes for Finishing Schemes for Ground and Ground Support Equipment
MIL-STD-810	Environmental Test Methods and Engineering Guidelines
MIL-STD-831	Test Reports, Preparation of
MIL-STD-889	Dissimilar Metals
MIL-STD-1186	Cushioning, Anchoring, Bracing, Blocking, and Waterproofing, with Appropriate Test Methods
MS33558	Numeral and Letters, Aircraft Instrument Dial, Standard Form of
MS33583	Tubing End, Double Flare, Standard Dimensions for
MS33584	Tubing End, Standard Dimensions for Flared
MS33649	Boss, Fluid Connection - Internal Straight Thread
MS33656	Fitting End, Standard Dimensions for Flared Tube Connection and Gasket Seal
MS33657	Fitting End, Standard Dimensions for Bulkhead Flared Tube Connection

AIR FORCE-NAVY AERONAUTICAL

AN929	Cap Assembly, Tube, Pressure Seal

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(Copies of specifications and standards required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer).

2.1.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN SOCIETY OF MECHANICAL ENGINEERS

Unfired Pressure Vessel Code

(Application for copies should be addressed to the American Society of Mechanical Engineers, United Engineering Center, 345 - 47th Street, New York NY 14304).

OFFICIAL CLASSIFICATION COMMITTEE

Uniform Freight Classification Rules

(Application for copies should be addressed to the official Classification Committee, Room 202, Union Station, 516 West Jackson Boulevard, Chicago, IL 60606).

NATIONAL FIRE PROTECTION ASSOCIATION NFPA 70 NATIONAL ELECTRICAL CODE

(Copies may be obtained from the National Fire Protection Association, 60 Batterymarch Street, Bostom, MA 02110).

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM B633 Electrodeposited Coatings of Zinc on Iron and Steel

ASTM D 3951 - Standard Practice for Commercial Packaging.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103).

SOCIETY OF AUTOMOTIVE ENGINEERS

AMS 5510 Steel Sheet, Strip, and Plate, Corrosion and Heat Resistant 18 CR-10.5 Ni-0.40 Ti (SAE 30321)

AMS 5512 Steel Sheet, Strip, and Plate, Corrosion and Heat Resistant 18 Cr-10.5 Ni-(Cb+Ta) (SAE 30347)

(Application for copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096).

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2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 First article. This specification makes provisions for first article inspection (see 4.4 and 6.3).

3.2 Components. The test stand shall consist of the following major components:

a. Cabinet (see 3.8.1)

b. Pneumatic system (see 3.8.2)

c. Electrical system (see 3.8.3)

3.3 Selection of specifications and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.4 Materials. Materials used in the construction of the test stand shall be as specified herein. Materials shall not be hammered, filled, nor otherwise treated to conceal defects. Materials not definitely specified shall be of uniform quality and condition and free of seams, cracks, or other defects that may adversely affect the appearance, strength, endurance, or wear resistance of the finished test stand.

3.4.1 Fungus proof materials. Materials that are nutrients for fungi shall not be used where it is practical to avoid them. Where used and not hermetically sealed, they shall be treated with fungicidal agents. However, if they will be used in a hermetically sealed enclosure, fungicidal treatment will not be necessary.

3.4.2 Protective treatment. When materials are used in the construction of the test stand that are subject to deterioration when exposed to climatic and environmental conditions likely to occur during service usage, they shall be protected against such deterioration in a manner that will in no way prevent compliance with the performance requirements of this specification. The use of any protective coating that will chip, crack, or scale with age or extremes of climatic and environmental conditions shall be avoided.

3.4.2.1 Reclaimed Materials. The use of reclaimed materials shall be encouraged to the maximum extent possible.

3.4.3 Metals. Metals shall be of the corrosion-resistant type or treated to resist corrosion due to salt spray or atmosphere conditions likely to be met in storage or normal service.

3.4.3.1 Dissimilar metals. The use of dissimilar metals, in contact as defined by MIL-STD-889, shall be avoided insofar as practicable. Metal plating or spraying of dissimilar base metals to provide suitable mating surfaces or separation by suitable insulating material shall be permissible.

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3.4.4 Standard parts. MS, AN, or MIL Standard parts shall be used wherever they are suitable for the purpose, and shall be identified by their part numbers. Commercial utility parts such as screws, bolts, nuts, fittings, et cetera, may be used, provided they have suitable properties and are replaceable by the MS, AN, or MIL Standard parts without alteration, and provided the corresponding MS, AN or MIL part numbers are referenced on the drawings, and in the parts lists. In applications for which no suitable corresponding MS, AN, or MIL part is in effect on date of invitation for bids, commercial parts may be used provided they conform to the requirements of this specification.

3.5 Design. The test stand shall be a semi-permanent test stand that requires a customer furnished source of 2000 ± 200 pounds-force per square inch nitrogen, and unless otherwise specified in the contract or order (see 6.2), 115 volt alternating current (AC), 60 cycle, single phase power. The stand shall provide pressures from a common source to four separate systems for testing pneumatic components with variable pressures from 0 to 5000 pounds-force per square inch (see Figure 1). Each pressure system shall have its own regulator, gauges, valves, and outlets. The four test systems shall be independently variable within their respective ranges of 0 to 5000 pounds-force per square inch, 0 to 3000 pounds-force per square inch, 0 to 600 pounds-force per square inch and 0 to 300 pounds-force per square inch. The 300 pounds-force per square inch system shall include a supplemental gauge for accurate indication of 0 to 60 pounds-force per square inch pressures. The pneumatic systems shall be arranged so that bottled nitrogen pressures can be used directly or after being pressurized to 6000 pounds-force per square inch. An electric timer, four separate flow meters, two manometers, a 12 volt direct current (DC) power supply, a 32 volt DC power supply and a test chamber shall be provided.

3.6 Construction. The test stand shall be so constructed that parts will not work loose in service. It shall be built to withstand the strains, jars, vibrations and other conditions incident to shipping, storage and service. Welding, riveting or other methods of assembly may be used providing subassemblies can be removed as required for service, maintenance, overhaul, inspection or replacement.

3.6.1 Locking devices. Screws, pins, bolts, et cetera, shall be furnished with standard locking devices where required. Safety wire, self locking nuts, cotterpins and lockwashers shall be acceptable.

3.6.2 Threads. All straight screw threads shall conform to the requirements of MIL-S-7742. Pipe threads where permitted herein shall conform to MIL-P-7105.

3.6.3 Welding. All welding shall conform to MIL-W-6858, MIL-W-8604 or MIL-W-8611, as applicable.

3.6.4 Adjustments and repairs. The stand shall be constructed so that simple hand tools may be used for maintenance and servicing. The requirements for special tools shall be held to a minimum.

3.6.5 Forces and loadings. The test stand shall be capable of withstanding the following load factors:

8.0 gravity (G) ultimate-forward and aft (lengthwise)
1.5G ultimate-side
4.5G ultimate-down
3.0G ultimate-lifting

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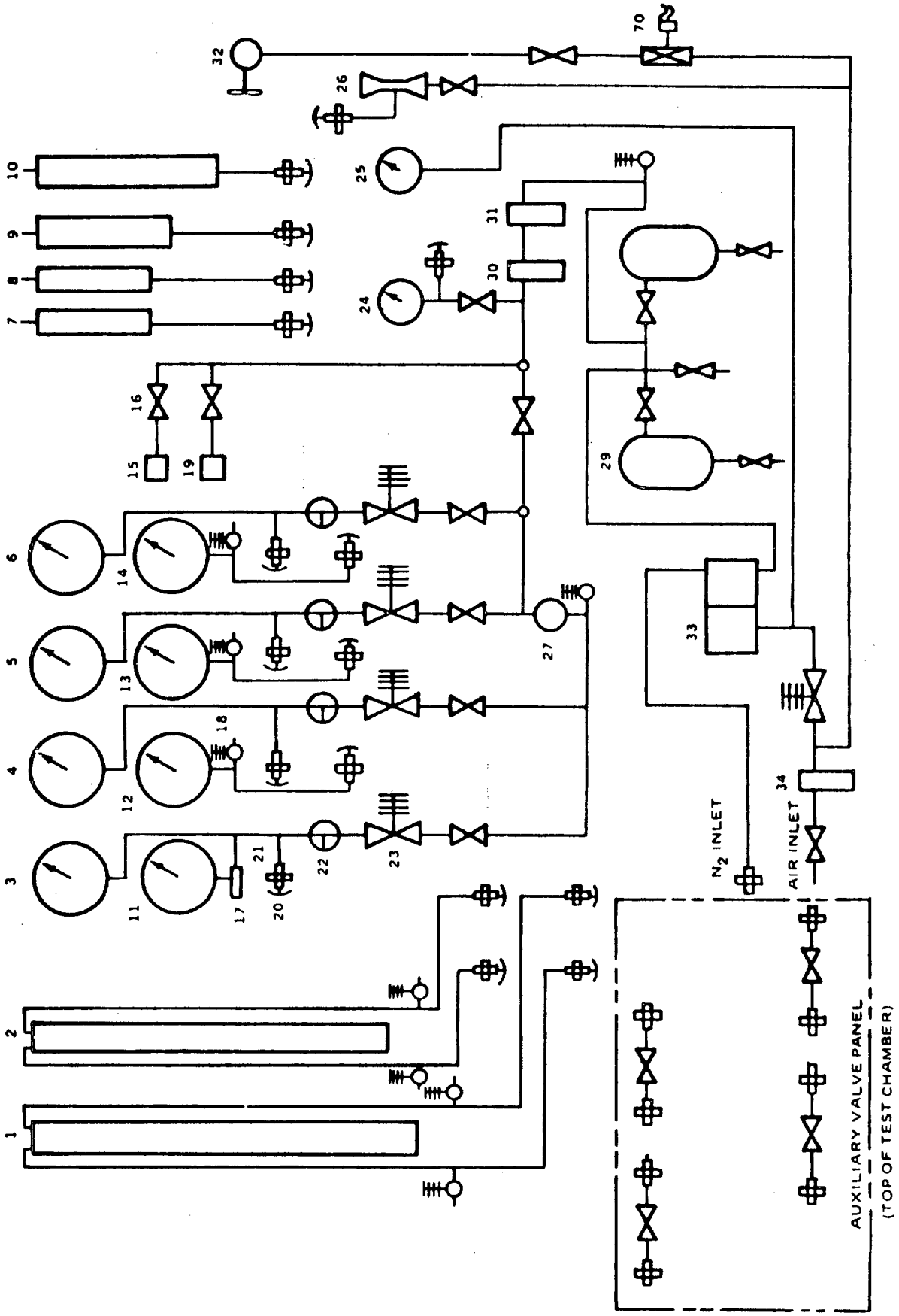


FIGURE 1. Pneumatic system flow diagram sheet

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LEGEND FOR PNEUMATIC SYSTEM BY MOUNTING LOCATION

	INSTRUMENT PANEL
1	MANOMETER (WATER)
2	MANOMETER (MERCURY)
3	PRESSURE GAGE (0-300 PSIG)
4	PRESSURE GAGE (0-600 PSIG)
5	PRESSURE GAGE (0-3000 PSIG)
6	PRESSURE GAGE (0-5000 PSIG)
7	FLOWMETER (271 SCCM MAX)
8	FLOWMETER (175 SCIM MAX)
9	FLOWMETER (1.30 SCFM MAX)
10	FLOWMETER (14.6 SCFM MAX)
11	PRESSURE GAGE (0-60 PSIG)
12	PRESSURE GAGE (0-200 PSIG)
13	PRESSURE GAGE (0-600 PSIG)
14	PRESSURE GAGE (0-3000 PSIG)
15	MOISTURE VAPOR INDICATOR
16	SHUTOFF VALVE (2 ON PANEL)
19	OIL VAPOR INDICATOR
	TEST PANEL (CHAMBER)
20	CAP (24)
21	BULKHEAD FITTING (24)
	a. REGULATED PRESSURE OUTLET - (4)
	b. AUX PRESSURE GAGE, MANOMETER, FLOWMETER AND VACUUM INLET - (12)
	c. AUX VALVE INLET & OUTLET - (8)
16	SHUTOFF VALVE (4)

	CONTROL PANEL
16	SHUTOFF VALVE (7)
	a. REGULATED PRESSURE SYSTEMS (4)
	b. VACUUM CONTROL (1)
	c. MAIN PRESSURE SYSTEM (1)
	d. SYSTEM PRESSURE GAGE (1)
20	BULKHEAD FITTING (1) TEST PT
21	CAP (1) FOR -24
22	SELECTOR VALVE (4)
23	PRESSURE REGULATOR (5)
	a. 0-300 PSIG
	b. 0-600 PSIG SYSTEM
	c. 0-3000 PSIG
	d. 0-5000 PSIG
	e. 0-125 PSIG - SHOP AIR
24	PRESSURE GAGE (0-6000 PSIG)
25	PRESSURE GAGE (0-160 PSIG)

	CABINET
26	EJECTOR PUMP
27	REGULATOR FIXED PRESSURE SET 6000 PSIG INLET/1000 PSIG OUT
70	SOLENOID VALVE
29	ACCUMULATORS (2)
30	FILTER 5 MICRON
31	PURIFIER (OIL/WATER SEPARATOR)
33	BOOST PUMP - 400 TO 6000 PSIG
34	FILTER/LUBRICATOR
17	GAGE SAVER
18	RELIEF VALVE (9)
16	SHUTOFF VALVE (6)
	a. BOOST PUMP BY-PASS (2)
	b. ACCUMULATOR DRAIN (2)
	c. AIR INLET
	d. SYSTEM VENT
	COUNTER TOP (OIL BATH)
32	AIR MOTOR
16	SHUTOFF VALVE

FIGURE 1. Pneumatic system flow diagram. (Cont'd)

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3.7 Performance.

3.7.1 Service Life. The stand shall be designed to be reliable and maintainable for a minimum useful operating life of 20 years.

3.7.1.1 Reliability. The test shall have a minimum Mean Time Between Failure (MTBF) of 250 hours. Failure is any malfunction which requires test stand "downtime" for repair, or replacement of any test stand component. Minor adjustments and replacement of filter and vapor indicator cartridges are exceptions.

3.7.1.2 Maintainability. The test stand shall be designed to assure maximum maintainability consistent with the requirements specified herein. The Mean Time to Repair (MTTR), assuming spare parts, tools, and test equipment are on hand, shall not exceed 2 hours. The **maximum** time required for any repair/replacement of any component or assembly shall not exceed 4 hours.

3.7.2 Capability. The stand shall be capable of testing pneumatic components requiring pressure range of 0 pounds-force per square inch to 5000 pounds-force per square inch with low air flows. The stand shall also be capable of testing pneumatic components requiring a pressure range of 0 to 3000 pounds-force per square inch with a maximum flow of 10.0 standard cubic feet per minute. The stand, by using a air ejector shall be capable of producing a vacuum of 23.4 inches of mercury. The stand shall be equipped with two DC power sources, one variable from 0 to 32 volts DC, and a second fixed source of 12 volts DC at 100 milliamperes. The stand shall also be capable of testing thermostats requiring an oil bath ranging in temperatures from 93° Fahrenheit (F) to 230°F (34°C to 110°C).

3.7.3 Environmental conditions.

3.7.3.1 Temperatures. The test stand shall be capable of satisfactory operations within the temperature range of 2° Centigrade (C) (35°F) to 51.7°C (125°F). It shall be capable of being stored in a dry (fluids drained) condition within the temperature range of -62°C (-80°F) to 71°C (+160°F) without detrimental effects.

3.7.3.2 Humidity. The test stand shall be capable of satisfactory operation in 95 percent relative humidity within the temperature range of 2°C (35°F) to 51.7°C (125°F) and satisfactory storage in a 100 percent relative humidity ambient within the temperature range of -54°C (-65°F) to 70°C (160°F) where condensation takes place in the form of water and frost.

3.7.3.3 Salt fog. The test stand shall be capable of satisfactory operation after exposure to a salt atmosphere as encountered in coastal areas.

3.7.3.4 Atmospheric pressure. The stand shall perform satisfactorily at full capacity from sea level to 6000 feet altitude.

3.7.3.5 Vibration. The test stand shall be capable of satisfactory operation after exposure to vibration incident to services use.

3.7.3.6 Fungus. The test stand shall be capable of satisfactory operation after exposure to fungus growth as encountered in tropical climates.

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3.7.3.7 Shock. The test stand shall be capable of withstanding the flight, taxiing and emergency landing loads specified in MIL-A-8421.

3.8 Details of components.

3.8.1 Cabinet. All components of the test stand shall be enclosed and protected by a cabinet. The cabinet shall have a neat, smooth exterior, free from sharp edges and corners. The cabinet shall consist of a welded steel or aluminum basic structure with integral flanges and welded to a suitable base that shall incorporate fork lift provisions. The cabinet shall be adequately braced to prevent distortion of excessive vibration during handling or operation of test stand.

3.8.1.1 Instrument panel. All electrical switches, gauges, flowmeters, manometers and other indicating devices except as noted in 3.8.1.2 shall be located on an instrument panel (see figure 2). The instrument panel shall have a colored line schematic on its face as an aid in operation of the stand's pneumatic system. A box recessed into the instrument panel shall be provided for storing the multimeter securely.

3.8.1.2 Control panel. All valves, regulators and gauges used to monitor the operation of the pneumatic system shall be located on a control panel, see Figure 2. The control panel shall have a colored line schematic on its face as an aid in operation of the stand's pneumatic system.

* 3.8.1.3 Counter top. The counter top shall be constructed of at least .063 inch thick 321 or 347 stainless steel conforming to AMS 5510 or AMS 5512 respectively. The counter top shall be attached to the cabinet with suitable hardware.

3.8.1.4 Test chamber. The test stand shall be equipped with a test chamber. The test chamber shall be located above the counter top and shall have effective inside dimensions of not less than 48 inches width by 14 inches depth by 16 inches height (see Figure 2). The test chamber shall include provisions for observing components under test and shall be designed and built to contain flying fragments from accidental bursts. The fragments from the bursts shall contain a maximum 1500 foot-pounds force of energy and shall be absorbed within the test chamber housing. Provisions shall be made to vent the test chamber.

3.8.1.4.1 Test chamber doors. The test chamber shall have two sliding doors across the front with an observation window in each door of approximately 12 inches by 12 inches effective viewing area. The observation window shall be of multiplate bullet resisting plate glass of glazing quality as defined by DD-G-451 and thickness of not less than 1½ inches. A third door shall be included on the outside end of the test chamber for loading components of extreme lengths and shall have an effective opening of not less than 14 inches wide by 16 inches high.

3.8.1.4.2 Illumination. The test chamber shall have a protected opening of 4 by 48 inches at the top to allow the light to shine through from the instrument panel. The protection shall be as specified for the observation window in 3.8.1.4.1.

3.8.1.5 Internal accessibility and ventilation. The cabinet shall be provided with sufficient access and ventilation doors and adjustable louvers to permit access for operating, servicing and ventilation of the test stand.

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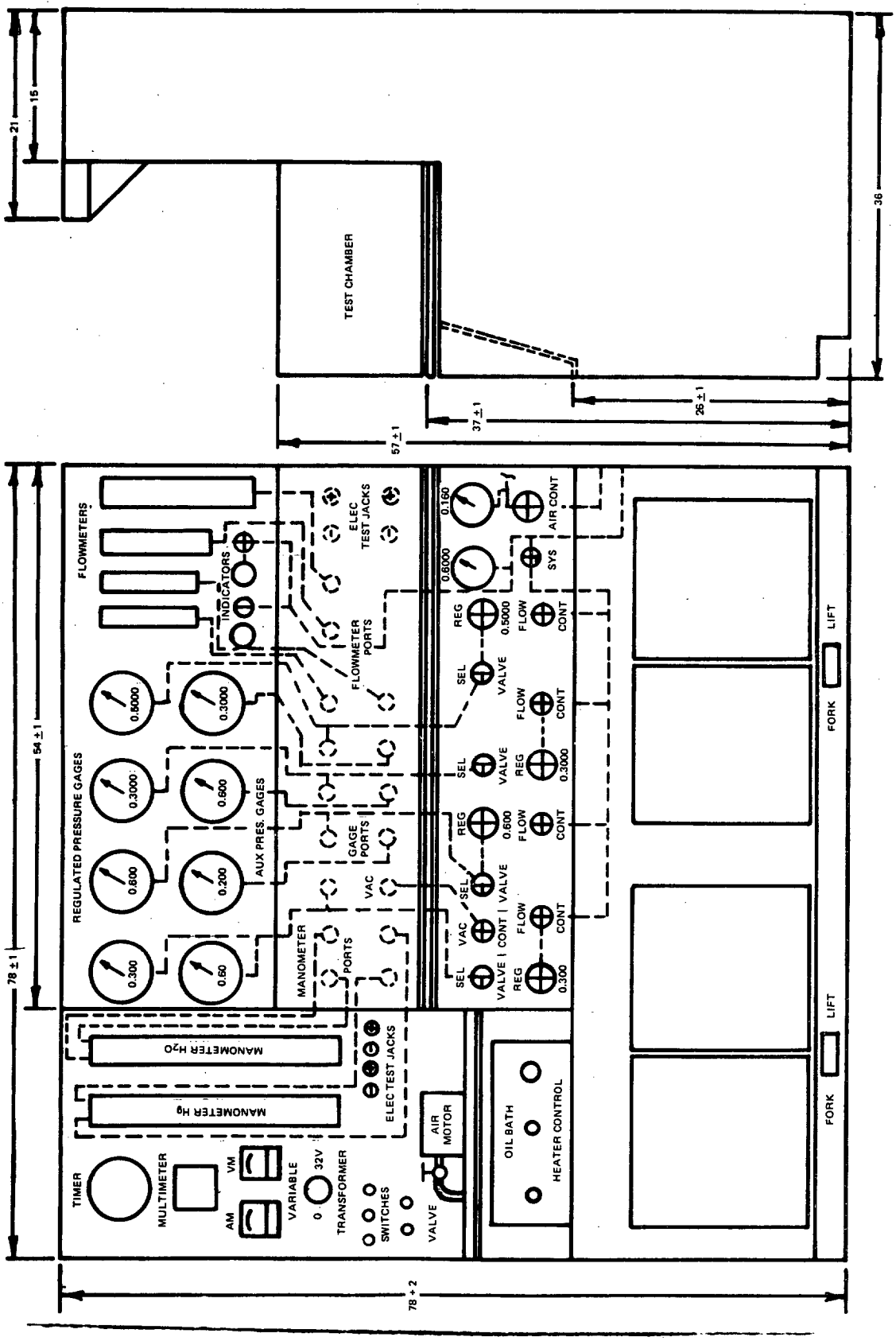


FIGURE 2. Test stand · component arrangement · envelope dimensions

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* 3.8.1.6 Doors. All doors shall provide a near flush exterior when the doors are in the closed position. Hidden type hinges shall be used. The rear of the cabinet shall have two doors each consisting of two 18 inch (approximate) door sections hinged together so that when the door is open, it will not extend more than 18 inches from the rear of the cabinet.

3.8.1.7 Compartments. A handbook compartment of 3 by 12 by 12 inches and an accessory storage box of 12 inches wide by 8 inches deep by 24 inches long shall be permanently attached at an accessible place within the cabinet.

3.8.1.8 Oil bath heater bowl. The oil bath heater bowl shall be constructed of stainless steel and shall have an approximate capacity of six quarts.

3.8.1.8.1 Oil bath temperature. The oil bath shall be capable of temperature from ambient to 230°F (110°C). Oil circulation shall be accomplished with an air motor operated stirrer. The temperature of the oil shall remain within 7°F (3.9°C) of the heater setting.

3.8.1.9 Immersion tank. A water immersion tank which can be placed inside the test chamber and stored below the counter top shall be provided. The tank shall be approximately 12 inches wide by 16 inches long and 12 inches deep. It shall be built so that the submerged component under test can be seen from the outside.

3.8.2 Pneumatic system. All components shall comply with the requirements of this specification. The test circuits shall be connected as shown in Figure 1 (the remaining portion of the diagram is furnished as information).

3.8.2.1 Source. The test stand shall operate from a customer furnished 2000 pounds-force per square inch bottled nitrogen source. The pneumatic systems, including the high pressure system, shall operate satisfactorily down to a source pressure of 500 $\pm$100 pounds-force per square inch.

3.8.2.2 Compressor. An air operated compressor shall be used to provide nitrogen supply pressures above source pressure. The compressor shall contain provisions for preventing the introduction of oil vapors into the nitrogen. The compressor shall operate on 90-125 pounds-force per square inch shop air requiring not more than 125 cubic feet per minute free air. The compressor shall contain a by-pass so that bottled nitrogen pressure can be used directly. The compressor shall be mounted on a base plate which will collect excess grease and moisture.

3.8.2.3 Accumulators. The high pressure accumulators shall be mounted vertically and equipped with drain valves for removing moisture from the accumulators without removing them from the test stand. They shall be of sufficient size to meet the pressure and flow requirements of this specification. The accumulators shall meet the requirements of the American Society of Mechanical Engineers Publications "Code for Unfired Pressure Vessels."

3.8.2.4 Test circuits. The test stand shall provide regulated test circuits for the following pressures:

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a. 0 to 300 pounds-force per square inch pressure

b. 0 to 600 pounds-force per square inch pressure

c. 0 to 3000 pounds-force per square inch pressure

d. 0 to 5000 pounds-force per square inch pressure.

3.8.2.4.1 Flow control and shutoff valve. Flow control shall be accomplished by metering valves upstream of the hand loading regulators. The stand shall be capable of controlling the following flow rates at the specified pressures for a minimum of 25 minutes:

	Test Stand Pressure System	Flow	Pressure
	0 to 300	100 standard cubic centimeters per minute	30.0 pounds-force per square inch
	0 to 300	170 standard cubic centimeters per minute	5.0 pounds-force per square inch
	0 to 300	4 standard cubic feet per minute	5.5 pounds-force per square inch
	0 to 300	4.5 standard cubic feet per minute	7.0 pounds-force per square inch
	0 to 300	13.5 standard cubic feet per minute	120 pounds-force per square inch
	0 to 300	14.0 standard cubic feet per minute	250.0 pounds-force per square inch
*	0 to 600	34.0 standard cubic inches per minute	180.0 pounds-force per square inch
	0 to 600	70.0 standard cubic inches per minute	245.0 pounds-force per square inch
	0 to 600	4.0 standard cubic feet per minute	5.5 pounds-force per square inch
	0 to 600	14.0 standard cubic feet per minute	250.0 pounds-force per square inch
	0 to 3000	6.5 standard cubic feet per minute	3000 pounds-force per square inch
	0 to 3000	10.0 standard cubic feet per minute	2500 pounds-force per square inch
	0 to 5000	6.0 standard cubic feet per minute	3000 pounds-force per square inch
	0 to 5000	9.5 standard cubic feet per minute.	2500 pounds-force per square inch.

3.8.2.4.2 Pressure regulators. Pressure regulation for each circuit shall be accomplished by separate single stage self-relieving hand loading pressure regulators. The pressure regulators shall be capable of maintaining 1 percent accuracy of the desired value.

3.8.2.4.3 Selector valves. There shall be a hand operated selector valve downstream of each hand loading regulator to allow for shutoff and bleeding of component under test.

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3.8.2.4.4 Gauges. High and low pressure gauges used in test circuits shall be provided on the instrument panel. The gauges shall be of solid front design with blowout backs. The gauges shall be class I, style A, size 6, type I, flush mounted with an accuracy of at least 1 percent of full scale range per GG-G-76. The gauges used to indicate system pressure (6000 pounds-force per square inch) and operating air pressure (0 to 160 pounds-force per square inch) shall be size 4 1/2, flush mounted conforming to GG-G-76. The gauges shall be mounted and connected as shown in Figures 1 and 2.

3.8.2.4.5 Gauge savers. The supplemental gauge in the 0 to 300 pounds-force per square inch test circuit shall be protected from pressures in excess of its rated pressure by a gauge saver. The auxiliary test gauges shall be protected by adequately rated relief valves set as follows:

a. 200 pounds-force per square inch - 200 $\pm$10

b. 600 pounds-force per square inch - 600 $\pm$25

c. 3000 pounds-force per square inch - 3000 $\pm$50

3.8.2.5 Flow Meters. The stand shall be equipped with four separate flow meters capable of measuring flows listed below. The flow meters shall be flush mounted and shall have an accuracy of at least 2 percent of maximum flow. Correction charts shall be furnished with each unit to permit correction of readings taken at temperature and pressures other than the standard used for calibration.

a. 271 standard cubic centimeter per minute maximum

b. 175 standard cubic inch per minute maximum

c. 1.30 standard cubic foot per minute maximum

d. 14.6 standard cubic foot per minute maximum

3.8.2.6 Manometers. The stand shall be equipped with a 24 inch water manometer calibrated in inches and tenths of inches of water and a 24 inch mercury manometer calibrated in inches and tenths of inches of mercury. The manometers shall be equipped with integral check valves to prevent manometer fluids from contaminating aircraft components under test.

3.8.2.7 Moisture vapor indicator. The test stand shall be equipped with a device for detecting the presence of moisture in the test media. The indicator shall be of the capsule type and indicate the presence of undesirable moisture vapor in the test media by changing color. The detecting capsule shall be located behind the instrument panel with provisions for viewing from the front and shall be easily removed and replaced. Provisions shall be included for safe bleeding of entrapped vapors.

3.8.2.8 Oil vapor indicator. The test stand shall be equipped with a device for detecting oil vapors in the test media. The oil indicator shall be of the capsule type and indicate the presence of undesirable oil vapors by changing color. The detecting capsule shall be located behind the instrument panel with provisions for viewing from the front and shall be easily removed and replaced. Provisions shall be included for safe bleeding of entrapped vapors.

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3.8.2.9 Ejector pump. The test stand shall be equipped with an ejector pump capable of creating a vacuum of 23.4 inches of mercury. Vacuum inlet shall be located within the test chamber and the air control valve shall be located on the control panel.

3.8.2.10 Auxiliary valve panel. The test stand shall be equipped with four control valves and fittings mounted on top of the test chamber for use in completing component test circuits. Valves shall be capable of meeting the maximum pressure and flow requirements of this specification.

3.8.2.11 Filters. A filter or filters capable of removing particles, water vapor and oil vapor shall be provided. The filters shall remove all oil vapors, remove all particles in excess of 5 micron absolute, and remove water vapor in excess of .25 grain of water vapor per pound of gas at 69°F and 14.7 pounds-force per square inch absolute pressure. The 5 micron filter shall be downstream of vapor filters to filter out any desiccant particles that might escape from vapor filters. The filters shall be capable of withstanding the normal operating pressures and also pressure drops of 0 to 200 pounds-force per square inch.

3.8.2.12 Fittings. Fitting ends shall be of the flared tube type in accordance with MS33656 and MS33657. Bosses shall be in accordance with MS33649. All test ports shall be equipped with caps conforming to AN929 attached to the port panel by metal cables.

3.8.2.13 Relief valves. Relief valves shall be provided in the system as necessary to protect test stand equipment from overpressurization.

3.8.2.14 Tubing. The pneumatic system tubing shall consist of 6061-T6 aluminum alloy tubing, MIL-T-7081 and 304 corrosion-resistant steel tubing, MIL-T-6845. Flaring shall be in accordance with MS33583 and MS33584.

* 3.8.3 Electrical system. The electrical system shall comply with the requirements as specified herein (see Figure 3) and the applicable requirements of the National Electrical Code NFPA 70. The test stand shall operate from a 60 cycle, single phase, 115 volts AC power source unless otherwise specified in the contract or order (see 6.2). The electrical systems of the test stands, that must operate from electrical power other than 60 cycle single phase, 115 volts AC power, may have added components to and/or different components than those of electrical systems of stands which operate from the 60 cycle, single phase, 115 volts AC power; however, such electrical systems shall comply with the requirements specified herein for end item performance.

3.8.3.1 Lighting. An overhead 48 inch fluorescent light shall be provided for instrument panel lighting. The flow meters shall be lighted by two 24 inch fluorescent lights installed behind the instrument panel. A translucent material between the lights and flowmeters shall be provided. Illumination shall be evenly distributed over the flowmeter surfaces and glare reduced to a minimum.

3.8.3.2 DC source. The test stand shall be equipped with two DC power supplies, one variable from 0 to 32 volts DC and a second power supply with a fixed output of 12 volts DC $\pm$0.05 percent at 100 milliamperes. Rectifiers shall be of the full wave bridge type and ripple voltage shall not exceed 3 percent. The 0 to 32 volts DC power supply shall have a continuous output

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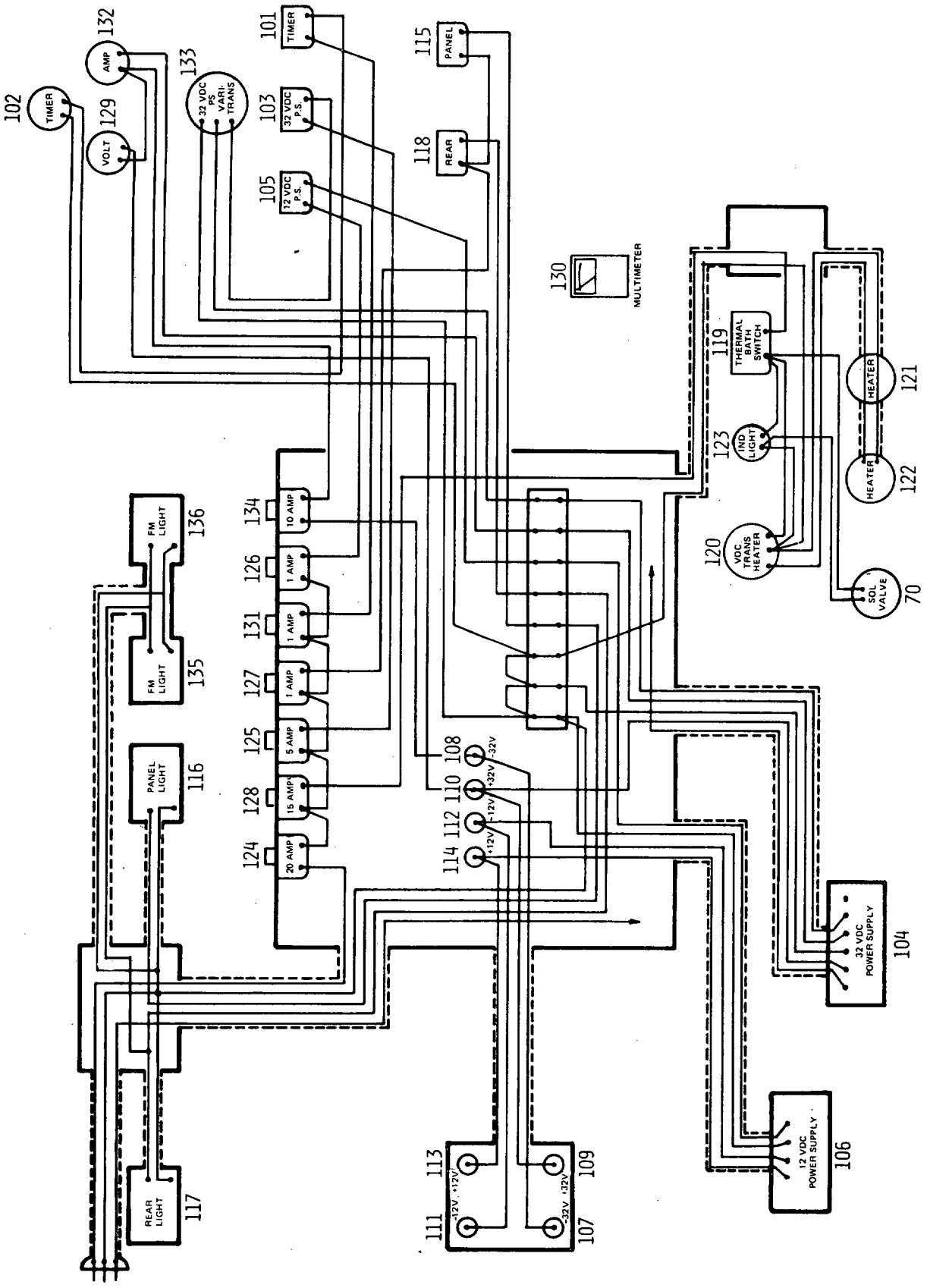


FIGURE 3. Electrical schematic

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SCHEMATIC NO.	ITEM DESCRIPTION
70	SOLENOID VALVE, ON-OFF FOR AIR MOTOR
101	SWITCH, ON-OFF, TIMER
102	TIMER, ELAPSED TIME, 0 TO 60 MINUTES
103	SWITCH, ON-OFF, 32 VDC POWER SUPPLY
104	POWER SUPPLY, 32 VDC
105	SWITCH, ON-OFF, 12 VDC POWER SUPPLY
106	POWER SUPPLY 12 VDC
107	TERMINAL, -32 VDC (CHAMBER)
108	TERMINAL, -32 VDC (PANEL)
109	TERMINAL, +32 VDC (CHAMBER)
110	TERMINAL, +32 VDC (PANEL)
111	TERMINAL, -12 VDC (CHAMBER)
112	TERMINAL, -12 VDC (PANEL)
113	TERMINAL, +12 VDC (CHAMBER)
114	TERMINAL, +12 VDC (PANEL)
115	SWITCH, ON-OFF, PANEL LIGHTS
116	LIGHT, PANEL, (48 INCH FLUORESCENT)
117	LIGHT, REAR CABINET INTERIOR (24 INCH FLUORESCENT)
118	SWITCH, ON-OFF, REAR/FLOWMETER LIGHTS
119	SWITCH, ON-OFF, OIL BATH HEATERS
120	CONTROL, VARIABLE VOLTAGE, HEATERS
121	HEATER, OIL BATH
122	HEATER, OIL BATH
123	LIGHT, ON-OFF, HEATER
124	CIRCUIT BREAKER, 20 AMP, ALL CIRCUITS
125	CIRCUIT BREAKER, 5 AMP, 32 VDC POWER SUPPLY
126	CIRCUIT BREAKER, 1 AMP, 12 VDC POWER SUPPLY
127	CIRCUIT BREAKER, 2 AMP, LIGHTING CIRCUITS
128	CIRCUIT BREAKER, 15 AMP; HEATER CIRCUIT
129	VOLTMETER, 0 TO 50 VOLTS
130	MULTIMETER, PORTABLE, AC & DC VOLTAGE, CURRENT, AND RESISTANCE MEASUREMENT
131	CIRCUIT BREAKER, 1 AMP, TIMER
132	AMMETER, 0 TO 10 AMPS
133	TRANSFORMER, VARIABLE, 32 VDC CONTROL
134	CIRCUIT BREAKER, 10 AMP, 32 VDC POWER SUPPLY OUTPUT
135	LIGHT, FLOWMETER (24 INCH FLUORESCENT)
136	LIGHT, FLOWMETER (24 INCH FLUORESCENT)

FIGURE 3. ELECTRICAL SCHEMATIC (CONT'D)

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of from 0 to 10 amperes and shall contain a panel mounted 0 to 10 amperes ammeter and a panel mounted 0 to 50 volts voltmeter. Meter accuracy shall be within 2 percent of full scale and no load to full load regulation of the power supply shall be within 15 percent.

3.8.3.3 Oil bath heater. The oil bath heater shall be equipped with two immersion heaters of 500 watt capacity minimum each. The immersion heater shall be controlled by a variable transformer of adequate rating. The oil bath heater control shall use an ON-OFF switch and ON-OFF light and the circuit breaker of adequate capacity. The heater shall be capable of temperatures from 93°F to 230°F (34°C to 110°C) and shall be able to maintain the desired temperature within 7°F (3.9°C). The heater shall be compatible with the following oils and fluids (or equivalents):

Cities Service Oil DC-4
Shell Tellus Oil No 72
Motor Oil SAE 20
Union Carbide Polyalkalene Glycol No 50-HB-280X

3.8.3.4 Power cable. An extra flexible, 3 conductor, 10 foot long cable with plug shall be permanently attached to the test stand. The plug shall have a nonmetallic body and cord grip, grounding and dead front construction as required by the National Electrical Code. The ground lead in the cable shall be mechanically grounded to the test stand.

3.8.3.5 Wire and wire identification. Stranded conductor of suitable degree of flexibility shall be used wherever practicable within the requirements of this specification. All wires shall be of such cross-section and temper as to provide ample and safe current carrying and mechanical strength. Where practical, wire shall not be smaller than American Wire Gage 22. Wires directly associated with the primary power circuit shall be of a size consistent with the ratings of the protective devices used. Insofar as practicable, all wiring within the test stand shall be distinctly numbered or color coded in order to facilitate testing and the location of faults.

3.8.3.6 Timer. A panel mounted electric elapsed time meter shall be provided. The timer shall have an RPM sweep second hand, a minute accumulation hand totaling 60 minutes, and a manual zero rest.

3.8.3.7 Multimeter. A portable multimeter with the minimum measurement capability, scale ranges and accuracies as shown on Table I shall be provided as part of the test stand. Sensitivity shall be 20,000 ohms/volt DC and 1000 ohms/volt AC minimum. The multimeter shall be provided complete with operator's manual, test leads and batteries. The multimeter shall be stored in the recessed box on the instrument panel of the test stand.

3.9 Interchangeability. All parts having the same manufacturer's part number shall be interchangeable with each other with respect to installation and performance with the exception of removable sheet metal doors and panels which may be furnished as replaceable spares, et cetera, with attaching holes to be drilled to match upon installation.

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3.10 Dimensions. The overall dimensions of the unit (see Figure 2) when ready for crating shall not exceed the following:

a. Length: 80 inches

b. Height: 80 inches

c. Depth: 36 inches

3.11 Weight. The weight shall be kept to a minimum consistent with strength requirements.

3.12 Color. The color of the finished test stand shall be aircraft gray conforming to TT-L-32, color number 16473 of FED-STD-595.

3.13 Finish.

3.13.1 Aluminum alloy parts. Aluminum surfaces may be finished painted in accordance with the provisions of MIL-STD-808, Code F-300, Type II.

3.13.2 Ferrous parts.

3.13.2.1 Paint finish. Surfaces of ferrous parts may be finish painted in accordance with the provisions of MIL-STD-808, Code F-100, Type II.

TABLE I - MEASUREMENT SCALE RANGE ACCURACY

MEASUREMENT	SCALE RANGE	ACCURACY
DC/AC Volts	0 to 2.5	±2% FSC (DC)/±3% FS (AC)
DC/AC Volts	0 to 10	±2% FSC (DC)/±3% FS (AC)
DC/AC Volts	0 to 50	±2% FSC (DC)/±3% FS (AC)
DC/AC Volts	0 to 250	±2% FSC (DC)/±3% FS (AC)
DC/AC Volts	0 to 1000	±2% FSC (DC)/±3% FS (AC)
DC/AC Volts	0 to 5000	±3% FSC (DC)/±4% FS (AC)
DC Microamperes	0 to 50	±1.5% FSC
DC Milliamperes	0 to 1	±2% FSC
DC Milliamperes	0 to 10	±2% FSC
DC Milliamperes	0 to 100	±2% FSC
DC Milliamperes	0 to 500	±2% FSC
DC Amperes	0 to 10	±2% FSC
Ohms	0 to 2000	±2.5° of Arc
Ohms	0 to 200K	±2.0° of Arc
Ohms	0 to 20 Meg	±2.0° of Arc

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3.13.2.2 Plated and coated parts. Surfaces of parts to be plates shall conform to the following specifications:

a. Chromium plating - QQ-C-320

b. Zinc plating - ASTM B633

c. Aluminum coating - MIL-C-83488

d. Zinc/chromate coating - MIL-C-87115

3.14 Marking and identification.

3.14.1 Stand marking. Each test stand shall be clearly and permanently identified by a securely attached nameplate in accordance with MIL-P-15024. Decals shall not be used. Serial numbers shall be assigned by manufacturer.

3.14.2 Part marking. All other parts shall be marked in accordance with MIL-STD-130.

3.14.3 Placards.

* 3.14.3.1 Instruction placards. All valves, gauges, outlets, connections and controls shall be identified by name and number. Placards shall have white machine engraved letters on a black background conforming to color number 17038 of FED-STD-595. Letters shall conform to MS33558. Instructional information on placards for valves and controls shall include OPEN, CLOSE, PRESSURE and VENT positions, and arrows pointing in proper direction for proper control, as applicable.

3.14.3.2 Warning placards. Warning placards shall have white, machine engraved letters on an insignia red background conforming to color number 11136 of FED-STD-595. Letters shall conform to MS33558.

3.14.4 Other marking. The exterior of the stand be stenciled to indicate fork lifts points and the following information on two sides "For air shipment this unit is designed for 8G Fore and Aft, 1.5G Sideward 4.5G Down and 3G Up." Letters shall be one inch high gloss black, color number 17038 of FED-STD-595.

3.15 Workmanship. The unit shall be fabricated and finished in a workmanlike manner. Particular attention shall be given to freedom from defects, burrs, sharp edges, accuracy of dimensions, radii of fillets, machine screw assemblies, welding, brazing and riveting, neatness and thoroughness of soldering, wiring, plating and marking of parts and assemblies.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of

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the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of tests. The inspection and testing of the test stand shall be classified as follows:

a.	First article inspection	See 4.4
b.	Quality conformance inspection	See 4.5

4.3 Test conditions.

4.3.1 Standard atmospheric conditions. Unless otherwise specified, all tests required by this specification shall be conducted at an atmospheric pressure of 29.5 to 30 inches mercury and temperatures of 65° to 75°F. (18.3°C to 23.9°C). When tests are conducted with atmospheric pressure or temperature differing from above values, correction shall be made for observed flows.

4.3.2 Test fluid. All tests shall be performed with dry air as defined in MIL-P-5518. Initial source pressure shall be 2200 $\pm$200 pounds-force per square inch.

4.3.3 Instrumentation. Instrumentation equipment used in conjunction with testing specified herein shall be of laboratory precision type and shall be calibrated at intervals of 20, 40, 50, 60, 75, 90 and 100 percent to insure laboratory accuracy.

4.3.4 Preliminary run-in and calibration. Preliminary run-in of the test stand as may be recommended by the manufacturer and calibration of gauges, meters, flow meters, manometers and power supplies shall be performed by the manufacturer prior to submission of the test stand for testing. A certified log of all preliminary running time and calibration data shall be prepared and submitted with test report.

4.4 First article inspections.

4.4.1 Test sample. The test sample shall consist of one test stand representative of the production equipment.

4.4.2 Test report. After the first article tests are completed, a test report prepared in accordance with MIL-STD-831 shall be furnished to the procuring activity.

4.4.2.1 Reliability information. The following information shall be incorporated as an attachment to the test report or shall be included as part of that report:

a. A certified log of all preliminary run-in time and calibration data.

b. A certified log of all failures, maintenance and adjustments identified by accumulated operating time, cycles or position in the test procedure.

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4.4.3 Inspections. The first article inspections shall consist of all the tests described under 4.6.

4.5 Quality conformance inspections. The quality conformance inspections shall consist of the individual tests.

4.5.1 Individual tests. Each test stand shall be subjected to the following tests as described under 4.6.

a. Examination of product	See 4.6.1
b. Pneumatic system tests	See 4.6.2
c. Electrical system tests	See 4.6.3

4.6 Test methods.

4.6.1 Examination of product. The test stand shall be inspected to determine compliance with the requirements specified herein with respect to parts, materials, dimensions, markings, finish, workmanship and ease of operation, maintenance and safety.

4.6.2 Pneumatic system tests.

4.6.2.1 Operation of 300 pounds-force per square inch system. With test stand connected to dry air source as specified in 4.3.2, connect a master test gauge between the 200 pounds-force per square inch auxiliary gauge through the auxiliary valve hook up and 300 pounds-force per square inch pressure outlets. Open selector and flow control valves and operate pressure regulator in increments of 10 pounds-force per square inch until relief valve on 200 pounds-force per square inch gauge opens. Close shutoff auxiliary valve to 200 pounds-force per square inch gauge and continue to operate pressure regulator to maximum range and back to zero. The readings observed on the 60, 200 and 300 pounds-force per square inch gauges shall compare with the pressure values on the master gauge within the accuracy limits specified in 3.8.2.4.4. Relief valve on 200 pounds-force per square inch gauge shall open at 200 $\pm$10 pounds-force per square inch.

4.6.2.2 Operation of 600 pounds-force per square inch system. With test stand connected to dry air source as specified in 4.3.2 except that initial source pressure may be below initial specified value, connect master test gauge and 600 pounds-force per square inch auxiliary gauge to 600 pounds-force per square inch pressure outlet. Open selector and flow control valves and operate pressure regulator in increments of 25 pounds-force per square inch until relief valve on 600 pounds-force per square inch gauges shall compare with the pressure values read on the master gauge within the accuracy limits specified in 3.8.2.4.4. Relief valve in auxiliary gauge shall open at 600 $\pm$25 pounds-force per square inch.

4.6.2.3 Operation of 3000 pounds-force per square inch system. With test stand connected to dry air source as specified in 4.3.2 except that source pressure may be below initial specified value, connect master test gauge and 3000 pounds-force per square inch auxiliary gauge to 3000 pounds-force per square inch pressure outlet. Operate compressor to raise source pressure to 3500 pounds-force per square inch and adjust compressor control valve so that compressor stalls when system pressure reaches 3500 $\pm$100 pounds-force

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per square inch. Open selector and flow valves and operate pressure regulator in increments of 500 pounds-force per square inch until relief valve on auxiliary 3000 pounds-force per square inch gauge opens and then return to zero. The readings observed on the stand's 3000 pounds-force per square inch gauges shall compare with the pressure values read on the master gauge within the accuracy limits specified in 3.8.2.4.4. Relief valve on auxiliary gauge shall open at 3000 ±50 pounds-force per square inch.

4.6.2.4 Operation of 5000 pounds-force per square inch system. With test stand connected to dry air source as specified in 4.3.2 and source pressure at 500 ±100 pounds-force per square inch, connect master test gauge to 5000 pounds-force per square inch pressure outlet. Operate compressor to raise source pressure to 5500 pounds-force per square inch and adjust compressor control valve so that compressor stalls when system pressure reaches 5500 ±100 pounds-force per square inch. Open selector and flow control valves and operate pressure regulator in increments of 500 pounds-force per square inch from 500 to 5000 pounds-force per square inch and back to zero pounds-force per square inch. The readings observed on the test stand 5000 pounds-force per square inch gauge shall compare with the pressure values read on the master gauge within the accuracy limits specified in 3.8.2.4.4.

4.6.2.5 Flow rate. The pressure systems shall be operated to demonstrate the flow requirements specified in 3.8.2.4.1. For the quality conformance inspection only, the flow demonstrations shall be for a minimum period of 2 minutes each. For these tests it will be necessary to include a control valve(s) from the auxiliary valve panel (3.8.2.10) in the flow circuit for flow control.

4.6.2.5.1 Flow meter correction charts. The accuracy of the charts shall be demonstrated by means of calculations and by demonstration with the collection of data. The data shall show the corrected readings for each of the flow meters.

4.6.2.6 Operation of ejector pump. Connect a different mercury manometer to "vacuum" test port and operate control valve slowly until manometer indicates at least 23.4 inches mercury vacuum. (3.8.2.9)

4.6.2.7 Relief valve function. Relief valves on auxiliary gauges which were not checked concurrently with tests in 4.6.2.2 and 4.6.2.3 due to system regulator limits shall be checked using next higher pressure system. Relief valves shall open at values specified in 3.8.2.4.5.

4.6.2.8 Leakage. With each pressure system outlet capped and valves opened, operate pressure regulator in each system and set pressure at mid-range. Pressure drop shall not exceed one percent of the full scale range of the respective gauge in thirty minutes.

4.6.2.9 Contamination test. Determine particle contamination from a 7 ±1 cubic feet sample of gas obtained from each pressure system. Particle contamination shall not exceed the following size and number distribution for each system:

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Size	Number of Particles
0-5 Micron	No limit
6-50 Micron	500
51-100 Micron	50
100-300 Micron	25
300 Micron and larger	0

4.6.3 Electrical system tests. The complete electrical system shall be subjected to the following tests at prevailing ambient conditions.

4.6.3.1 Continuity tests. A complete continuity check shall be made to determine compliance with the electrical diagram.

4.6.3.2 DC source. The test stand shall be checked for compliance with requirements of 3.8.3.2.

4.6.3.3 Oil bath heater. The test stand shall be checked for compliance with requirements of 3.8.3.3. Any one of the fluids or oils listed may be used for the tests.

4.6.4 Serviceability. The complete test stand shall be inspected and evaluated for ease of maintenance, service, adjustment of controls and accessibility.

4.6.5 Acceleration loads. In lieu of the tests specified in MIL-A-8421 a static stress analysis of the test stand shall be submitted.

4.6.6 Environmental tests. The test stand shall be subjected to the following tests conducted in accordance with the specified procedures of MIL-STD-810. Upon completion of each test, the test stand shall be subjected to the tests specified in 4.6.2. Failure of the test stand to operate and meet the accuracies specified during or after each exposure, failure of any item or assembly due to deterioration from exposure in these tests, or formation of any undesirable condition that cannot be remedied by routine maintenance shall be cause for rejection.

4.6.6.1 Prior to conducting any of the tests specified herein, the equipment shall be operated under standard conditions and a record made of all data necessary to provide criteria for checking satisfactory performance of the equipment during or after environmental tests.

4.6.6.2 High temperature test (storage). The stand shall be placed within a high temperature chamber and the chamber temperature raised to 71°C (160°F) with an internal relative humidity of not more than 15 percent for a period of 48 hours. The temperature shall then be reduced to standard conditions, retested, and results compared with 4.6.6.1.

4.6.6.3 Low temperature test (storage). For this test the flow meter tubes and floats shall be removed and packed in a suitable protective container and placed inside the storage area of the test stand. The stand shall be placed in a low temperature chamber and the chamber cooled to and maintained at a temperature of -62°C (-80°F) for a period of 72 hours. At the conclusion of this exposure period, the temperature shall be raised to prevailing room temperature. The flowmeters shall be reassembled. The stand shall then be operated and the results compared with 4.6.6.1.

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4.6.6.4 Humidity test. The stand shall be placed in a test chamber and set up to simulate installed conditions. Provision shall be made to prevent moisture dripping on the stand from above. The stand shall be subjected to a relative humidity of 92 ±2 percent at a room temperature of 80°F to 85°F (26.7°C to 29.4°C) for a period of 20 hours. The stand shall be operated at the conclusion of this test and the data compared with 4.6.6.1.

4.6.6.5 Salt-fog. The test stand shall be subjected to salt-fog in accordance with method 509, procedure I for 50 hours.

4.6.6.6 Fungus resistance. The test stand shall be subjected to fungus in accordance with method 508, procedure I. In lieu of fungus test the contractor may submit certification that fungus-nutrient materials have been treated in accordance with the provisions of MIL-T-152, Type 2 treatment.

4.6.6.7 Vibration. The test stand shall be subjected to vibration tests upon completion of all tests to determine that no damage has resulted. The tests shall be in accordance with Method 514.3, equipment category 1 of MIL-STD-810D.

4.6.7 Service life. Service life tests as specified in 4.6.7.1, 4.6.7.2, 4.6.7.3 and 4.6.7.4 shall be accomplished on the test stand. Data, as defined in 4.4.2.1b, shall be recorded and used to show whether the test stand meets the specified reliability (3.7.1.1) and maintainability (3.7.1.2) requirements.

NOTE: Where the operations described below are accomplished as portions of other tests described here, they shall be counted as part of the service life tests.

4.6.7.1 Pneumatic system pressure tests. The 180 cycles of operation, as defined herein, shall be accomplished on the test stand. The test stand systems shall be pressurized as specified in 4.6.2.1, 4.6.2.2, 4.6.2.3, and 4.6.2.4.

4.6.7.2 Leakage. At completion of tests specified above, perform leakage test in accordance with 4.6.2.8.

4.6.7.3 DC source. The dc power sources shall be operated for 200 hours at values specified in 3.8.3.2.

4.6.7.4 Oil bath heater. The oil bath heater shall be operated for 200 hours at a temperature of 215°F to 230°F (102°C to 110°C). The air motor operated stirrer shall continuously circulate the oil during this test.

5. PACKAGING

* 5.1 Preservation. Preservation shall be Level A or commercial, as specified (see 6.2).

* 5.1.1 Level A. Each test stand shall be preserved and packaged in accordance with Method IIA of MIL-P-116. All openings of the test stand and pneumatic system shall be plugged, capped or covered with barrier material conforming to MIL-B-121, Grade A. When MIL-B-121 barrier material is used, it shall be secured in place with tape conforming to MIL-T-22085.

* 5.1.2 Commercial. Test stands shall be preserved and packaged in accordance with ASTM-D-3951.

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5.2 Packing. Packing shall be Level A, B or C or commercial, as specified (see 6.2).

5.2.1 Level A and B. Each test stand preserved and packaged as specified in 5.1.1 shall be packed in a container conforming to MIL-C-104, Type I, Class 2. Containers shall be of uniform size and shape and of minimum tare and cube. Closure and strapping shall be in accordance with MIL-C-104.

* 5.2.2 Commercial. Test stands preserved in 5.1.2 shall be packed in accordance with ASTM-D-3951.

5.3 Physical protection. Interior cushioning, anchoring, blocking and bracing shall be in accordance with MIL-STD-1186. Tests: Edgewise-drop test or cornerwise drop test shall be in accordance with Appendix A of the specification. Tests are not required when Level C packing is specified.

5.4 Marking. In addition to any special marking required by the contract or order, interior packages and exterior containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The unit covered by this specification is intended for testing individual aircraft pneumatic components after repair and to functionally test individual components in the event of a pneumatic system malfunction.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification.

b. Selection of applicable level of preservation, packaging, and packing (see 5.1 and 5.2).

c. The electrical power the test stand will be connected to for operation if other than 60 cycles, single phase, 115 volts AC power (see 3.8.3).

* 6.3 First article. When a first article inspection is required, the item will be tested and should be a standard production item as specified in 4.4.1. The first article should consist of (one) unit(s). The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, test approval of the documents first article.

6.4 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Air Force - 99
Army - AV
Navy - AS

Preparing Activity:
Air Force - 82

(Project 4920-0397)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-T-83773B	Test Stand Aircraft Pneumatic Components

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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